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ECE 308 Assignment 3
Implementing the Cascade Control System

1) Designing Control Parameters

Known values:

a = 40.323
b = 61.694;
wcf = 100 * pi;
zcf = 0.9;
Kp = 13.6;
Kv = -0.01
Kbb = 0.419

For 10% Overshoot:

Kc = 4.773;
z = 1.431
outer wn^2 = kbb * kc * z ➔ outer wn = 1.692
zeta = KbbKc/(2wn) = (0.419 * 4.773)/(2 * 1.692) = 0.59

Calculating Kc and z for ≤ 8% PO and 0.2s ts:

e^[-zeta * pi / (1-zeta^2)^(0.5)] = 0.08
zeta = -ln(0.08)/(pi^2 + ln(0.08)^2)^(0.5) = 0.626
Kc = 2 * zeta * wn / Kbb = 2 * 0.626 * 1.692 / (0.419) = 5.06
z = wn^2/(Kbb * Kc) = 1.692^2/(0.419 * 5.06) = 1.35

Final Kc and Z values:

Kc = 5.06; z = 1.35; all other values unchanged .

2) Simulink Diagrams

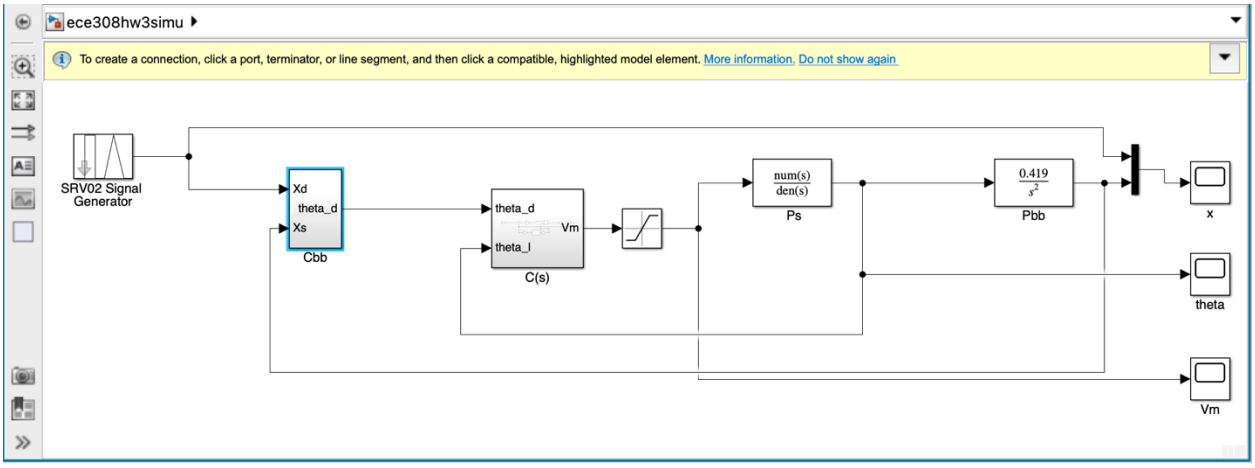


Figure 1: Simulink Cascade PID System

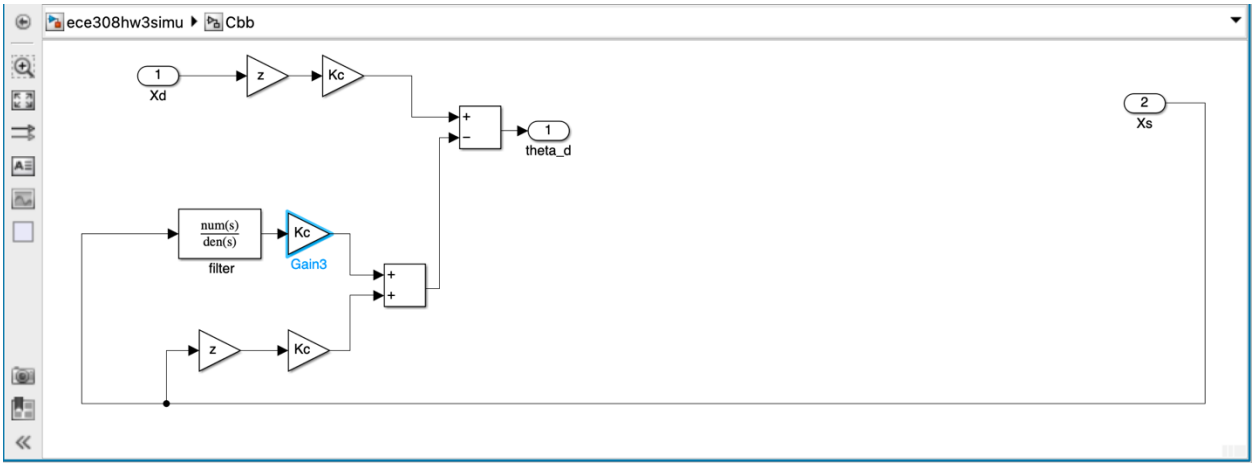


Figure 2: Cbb

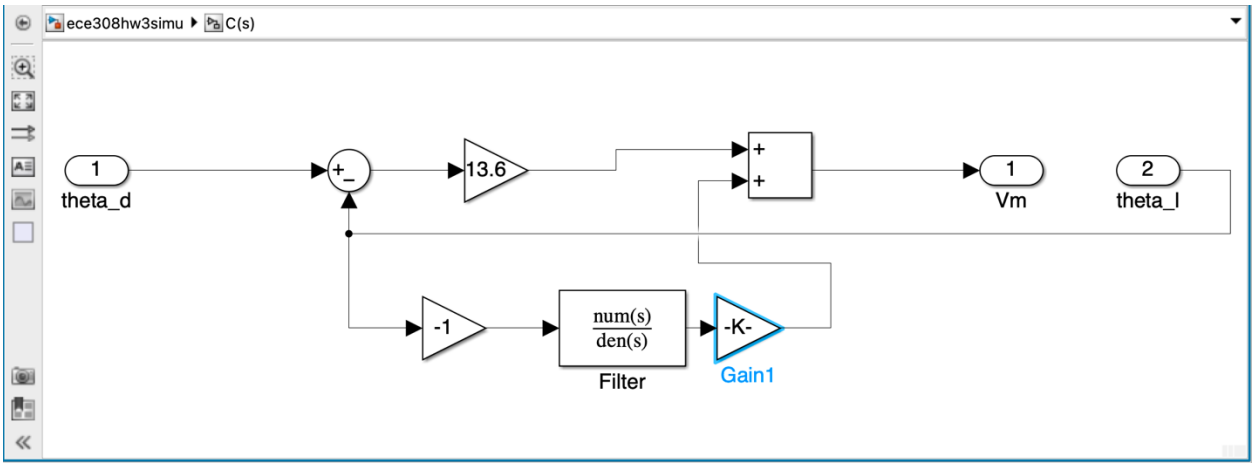


Figure 3: Cs

3) Measurements and Requirements

Steady State Value: 0.05m
Peak Value: 0.0529m
Max Overshoot: 5.8%
Peak Time: 1.94s
Settling Time: 2.396s
Rise Time: 1.56s

The designed PID controller met specifications well, the overshoot was underneath the 8% maximum PO. The steady state value matched the intended signal exactly. The only problem was the rise time was significantly higher than its maximum value, which in theory should be able to be shrunk if Kc or z was made smaller.

4) Plots

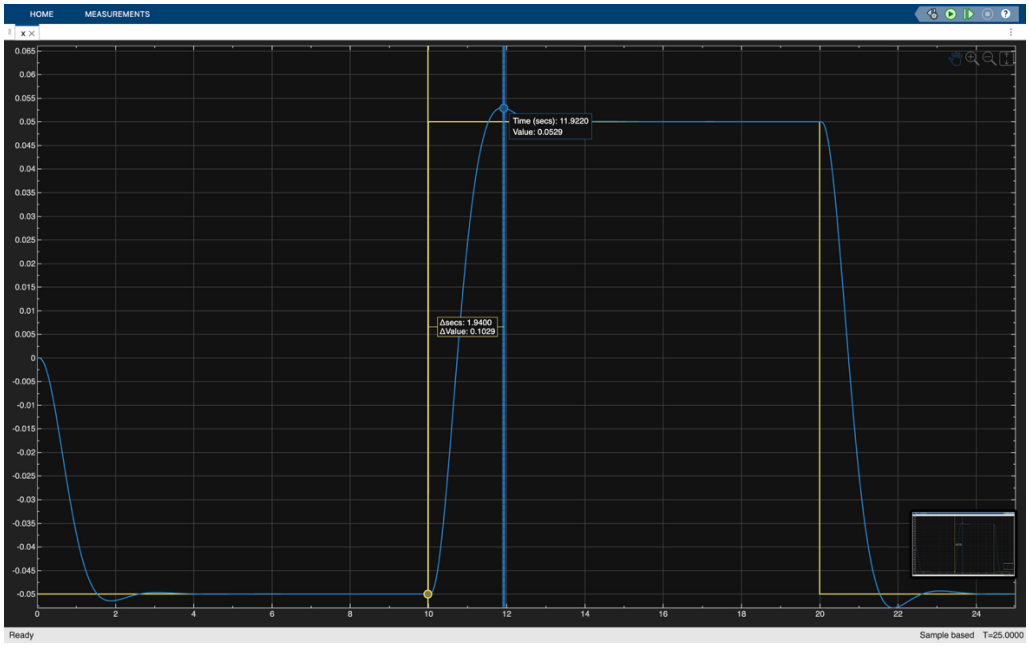


Figure 4: x(m) Measured for Peak Time and Percentage Overshoot

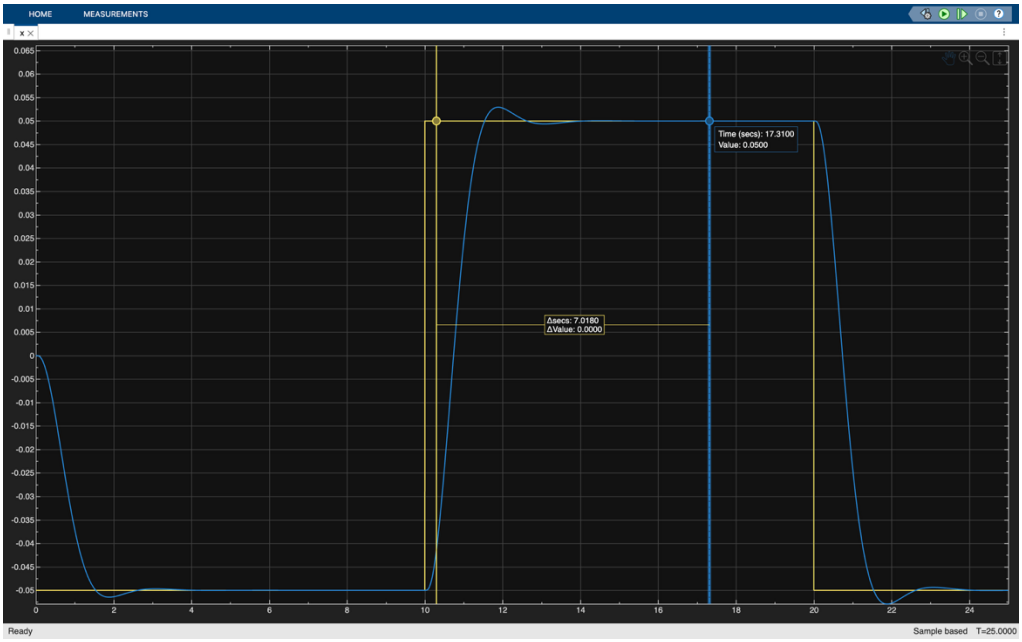


Figure 5: x(m) Measured for Steady Sate Response

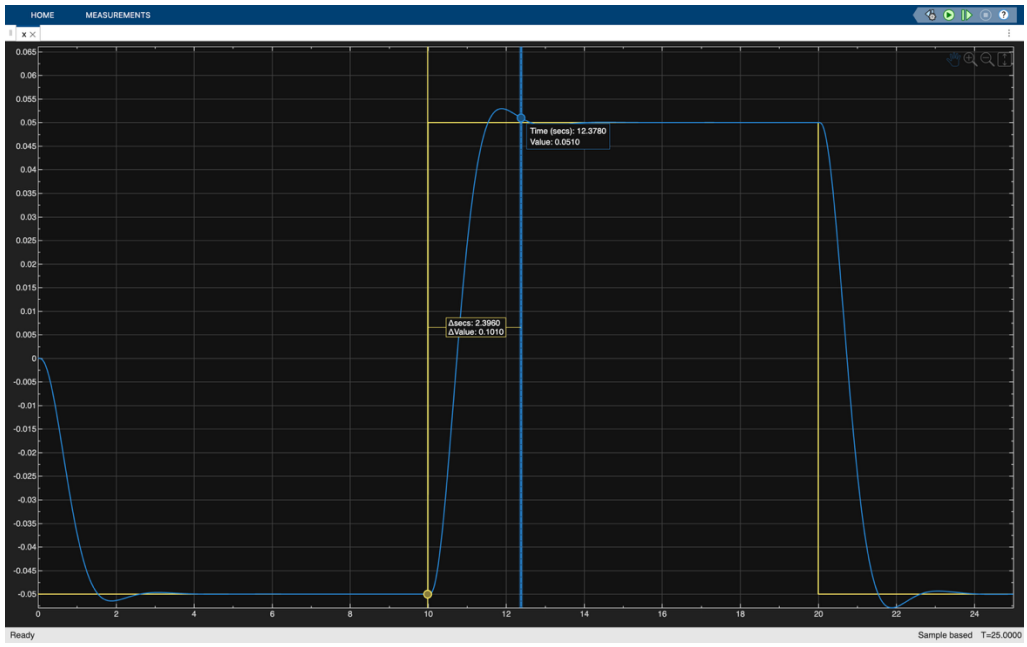


Figure 6: x(m) Measured for Settling Time

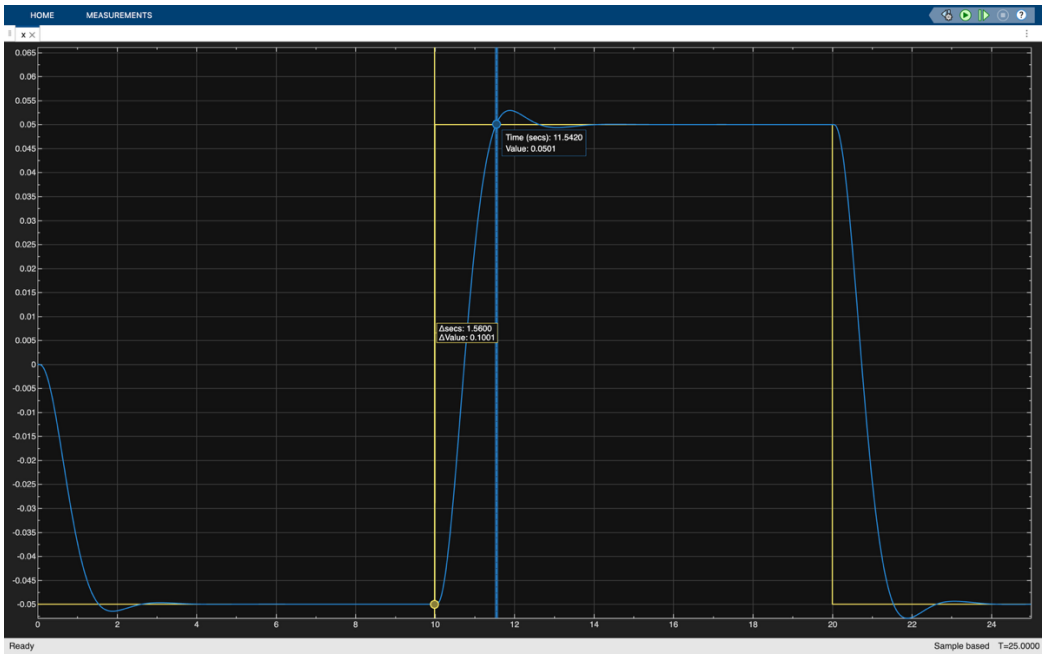


Figure 7: x(m) Measured for Rise Time

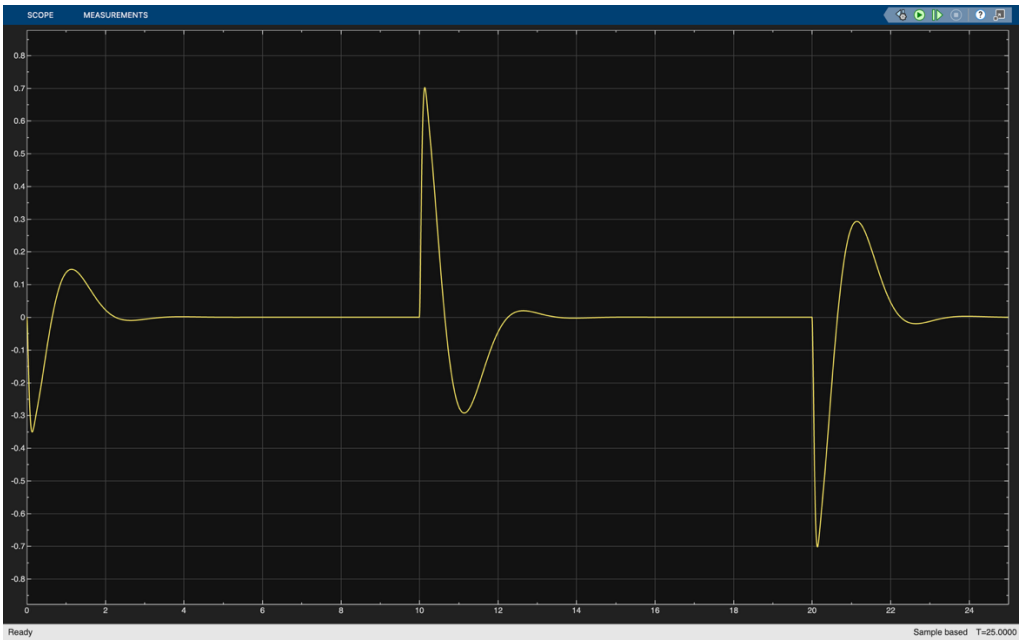


Figure 8: Angle Theta_d (deg)

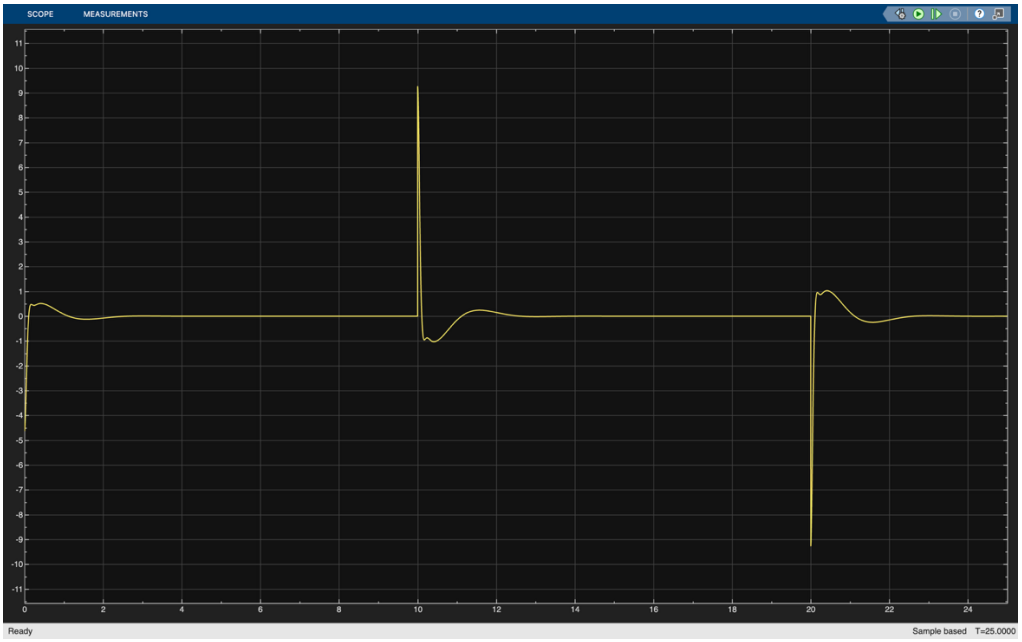


Figure 9: Applied Voltage (V)